



COURSE DESCRIPTION FORM

INSTITUTION

FAST School of Computing, National University
of Computer and Emerging Sciences, Islamabad

PROGRAM (S) TO BE EVALUATED

MSDS-02
Spring 2026

A. Course Description

This course provides an advanced study of the theoretical foundations, practical considerations, and research challenges involved in training and deploying deep neural networks. It examines core architectural principles and learning mechanisms underlying modern deep learning models, with an emphasis on gradient-based optimization and representation learning. The course explores a range of deep neural network architectures and learning paradigms used in contemporary research and applications, including convolutional, recurrent, generative, and reinforcement learning-based models. Students will engage with current research literature to understand model design choices, training dynamics, evaluation methodologies, and limitations. The course also addresses advanced topics such as generative modeling, sequential learning, and decision-making using deep neural networks, preparing students for research-oriented and applied work in deep learning.

Course Code	DS-5006		
Course Title	Deep Learning		
Credit Hours	3		
Prerequisites by Course(s) and Topics	None		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Final Exam:	40	
	Sessional 1:	10	
	Sessional 2:	10	
	Assignments:	10	
	Quizzes:	10	
	Project:	20	
	Explanation of Assessment: All assignments and project deliverables must be submitted by the stated deadline. Late submissions will not be accepted and will receive a grade of zero. Students are responsible for verifying their submissions by downloading and carefully reviewing all submitted deliverables before the deadline.		



Course Coordinator	Dr. Zohair Ahmed										
Course Instructor	Dr. Zohair Ahmed										
URL (if any)	-										
Current Catalog Description	This course provides an advanced treatment of the theory, applications, and practical challenges involved in training and deploying deep neural networks. It examines fundamental building blocks, architectural design principles, and gradient-based optimization techniques underlying modern deep learning systems. The course explores a range of neural network architectures and learning paradigms used in contemporary research, including convolutional, recurrent, generative, and reinforcement learning-based models. Emphasis is placed on representation learning, model evaluation, training dynamics, and scalability considerations. Students engage with current research literature to critically analyze state-of-the-art methods and understand their limitations, preparing them for advanced research and applied work in deep learning.										
Textbook (or Laboratory Manual for Laboratory Courses)	1. Deep Learning, by Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press. Available Online										
Reference Material	2. Neural Networks and Deep Learning: A Textbook, Charu C. Aggarwal 3. Deep Learning with Python, François Chollet 4. Machine Learning and Deep Learning with Python, scikit-learn and TensorFlow, Sebastian Raschka and Vahid Mirjalili										
Resources from Internet											
Innovative Teaching Methods	Use of Multimedia systems, Collaborative Learning (group projects), Ethical AI and Social Impact Discussions, Case-Based Learning.										
Course Goals	<table border="1"><thead><tr><th colspan="2">Course Learning Outcomes (CLOs)</th></tr></thead><tbody><tr><td>CLO 1</td><td>Analyze and evaluate the theoretical foundations, architectures, and optimization principles of deep neural networks.</td></tr><tr><td>CLO 2</td><td>Design, adapt, and assess deep learning models for complex data and problem settings while considering generalization, robustness, and computational constraints.</td></tr><tr><td>CLO 3</td><td>Critically evaluate advanced deep learning architectures and learning paradigms across diverse application domains.</td></tr><tr><td>CLO 4</td><td>Interpret, critique, and synthesize contemporary research literature in deep learning to identify emerging trends, limitations, and research opportunities.</td></tr></tbody></table>	Course Learning Outcomes (CLOs)		CLO 1	Analyze and evaluate the theoretical foundations, architectures, and optimization principles of deep neural networks.	CLO 2	Design, adapt, and assess deep learning models for complex data and problem settings while considering generalization, robustness, and computational constraints.	CLO 3	Critically evaluate advanced deep learning architectures and learning paradigms across diverse application domains.	CLO 4	Interpret, critique, and synthesize contemporary research literature in deep learning to identify emerging trends, limitations, and research opportunities.
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Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Week #	List of Topics	Hrs	CLOs	Assessments
	Week 1	Introduction • Introduction to the Course • Why Learning is essential? • What is Artificial Intelligence, Machine Learning, and Deep Learning? • Learning vs Inference – Overfitting and Generalization • Types of Learning: Fully Supervised, Unsupervised, Semi-supervised, Self Supervised, and Reinforcement Learning. Regression vs Classification. • ChatGPT and Large Language Models • What is Intelligence? perspective on intelligence • The anticipated world out of AI • Embracing AI or Resisting AI – Role of Natural Selection • Real-world applications from numerous domains.	3	1	
	Week 2	Machine Learning Overview • Models for Linear Regression • Models for Linear Classification Linear Regression • Choosing Regression Line, Error Function/Loss Functions, • Hypothesis, Derivatives • Optimization Algorithms, Gradient Descent • Model Selection	3	1, 2	Assignment #1
	Week 3	Deep Learning – A Modern Learning Paradigm • Perceptron and Multi-layered Perceptron model. • When is Learning possible? • (Multi-layered) Sigmoid neuron model. • Neural network architecture and MNIST problem. • Backpropagation	3	1, 2	
	Week 4-5	Aspects of Neural Network • Activation functions. • Vanishing gradient problem and exploding gradient problem. • Initialization schemes. • Overfitting, underfitting, and reasonable fitting in deep neural networks. • Regularization includes L1 and L2 weight decay, dropout, data augmentation, and early stopping. • Implicit vs Explicit Regularization. • Bias-Variance trade-off, Model complexity, and Training-Validation performance trade-off. • Batch size.	3	1, 2	

	Week 6	Sessional -I	Hrs	CLOs	Assessments
	Week 7	Introduction to Deep Convolutional Neural • LeNet, AlexNet, VGGNet	3	3	Select Project Topic
	Week 8	Deep Learning for Object Detection and Image Segmentation – • YOLO (You Only Look Once)	3	1,2,3	Assignment #2
	Week 9	Deep Learning for Object Detection and Image Segmentation: • SSD (Single Shot MultiBox Detector) • R-CNN Family	3	1,2,3	
	Week 10	Introduction to Semantic segmentation • Well known architectures: U-Net, ResNet, • DenseNet, EfficientNet	3	1,2,3	Assignment #3
	Week 11	Introduction to Transformers – • Vision Transformers (ViT)	3	3	
	Week 12	Sessional - II			
	Week 13	Introduction to Graph Neural Networks • Machine Learning on Graphs • Graph Convolution Filters	3	3	
	Week 14	Deep Reinforcement Learning • Q-learning • Policy grad	3	3	
	Week 15	Explainability in Deep Learning • Importance of model interpretability • Techniques: SHAP, LIME, Grad-CAM	3	3	Project Submission
		Final Exam	48		
Laboratory Projects/Experiments Done in the Course	Multiple architectures' implementation will be discussed in class and students will be required to do one project in which a paper will be implemented				
Programming Assignments Done in the Course	Sr#	Tool Name	Course Topics in which tool is used		
	1	Colab, Jupiter notbook, Matlab, Knime	Implement all ML models like linear Classification, Linear Regression, Nonlinear, classification, Kernel methods, SVMs, Neural Networks etc		



Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	25	10	8	2
Oral and Written Communications	Every student is required to submit at least __1__ written project reports of typically __5__ pages and to make __1__ oral presentations of typically __15__ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

Instructor Name: Dr. Zohair Ahmed

Instructor Signature _____

Date: 19^h January 2026